

Hunting Invisible Game

By: Gage Mariano

What is a deepfake?

deepfake noun

deep·fake

'dēp-, fāk ▶▶

plural deepfakes

: an image or recording that has been convincingly altered and manipulated to misrepresent someone as doing or saying something that was not actually done or said

Real or FAKE?



Can you spot the REAL image?



Problem Motivation



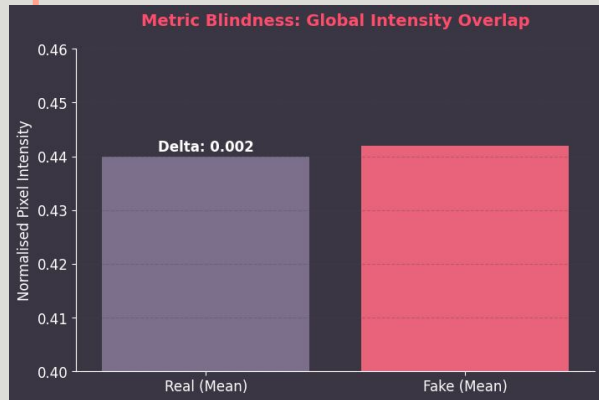
Why is detecting deepfakes important?

- The loss of objective reality:
 - Disinformation
 - Fraud
- Privacy
- Accessibility



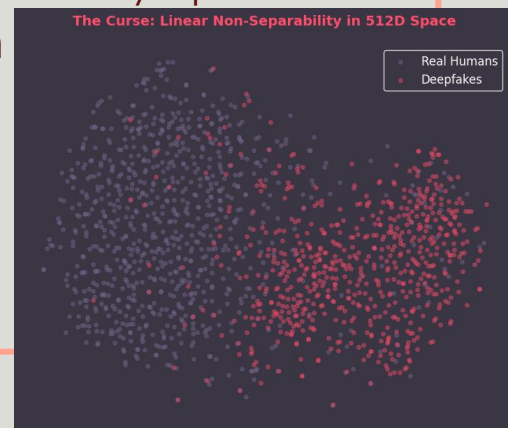
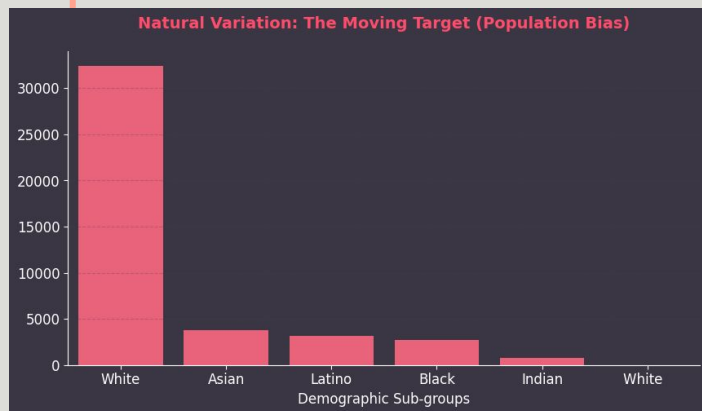
Approach Motivation

6



Why is detection so difficult?

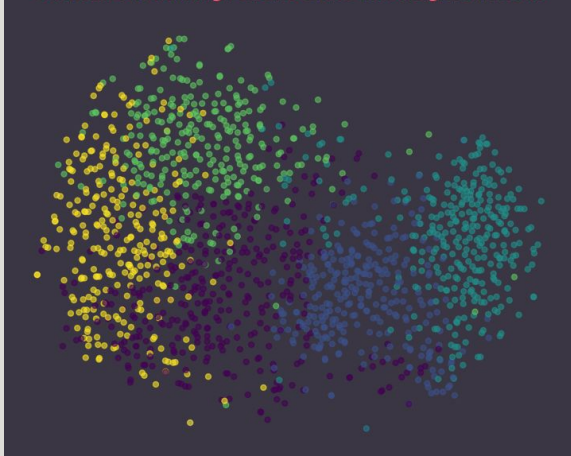
- Curse of dimensionality
- Natural variations
 - Which most datasets do not uniformly represent
- No rule of thumb solution
- Adversarial nature



Proposed Solutions:

How do we overcome detection difficulties?

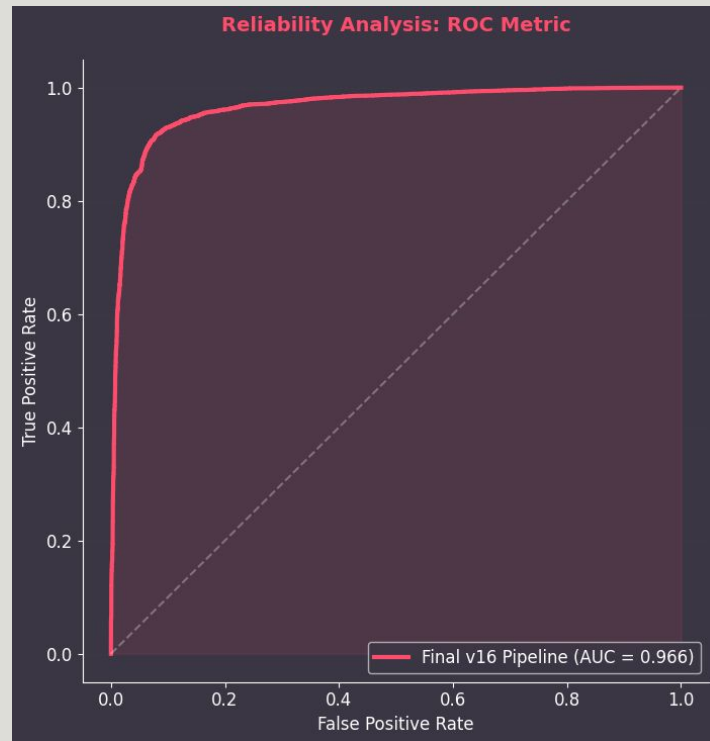
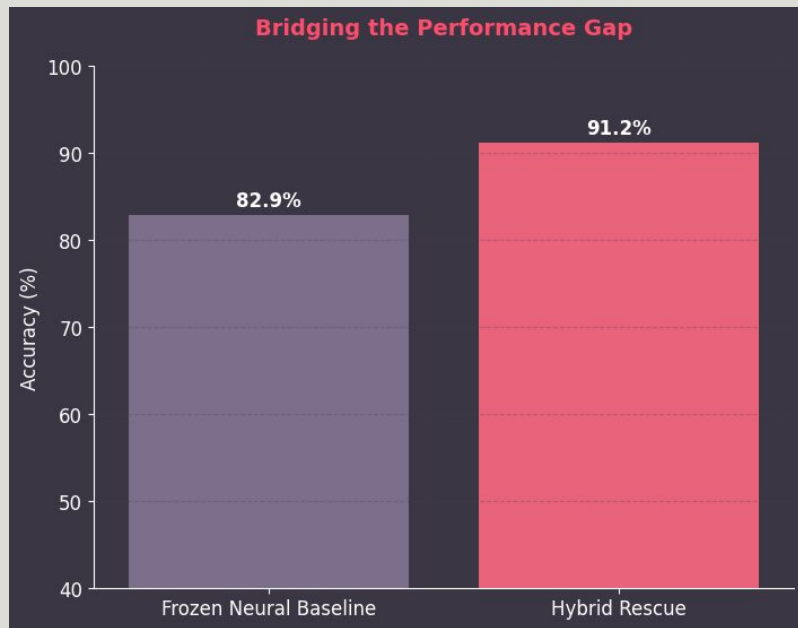
Solution: Partitioning Feature Chaos into Neighbourhoods



- Encoder (ResNet-18)
- Clustering (K-Means)
- Anomaly Detection (Isolation Forest)

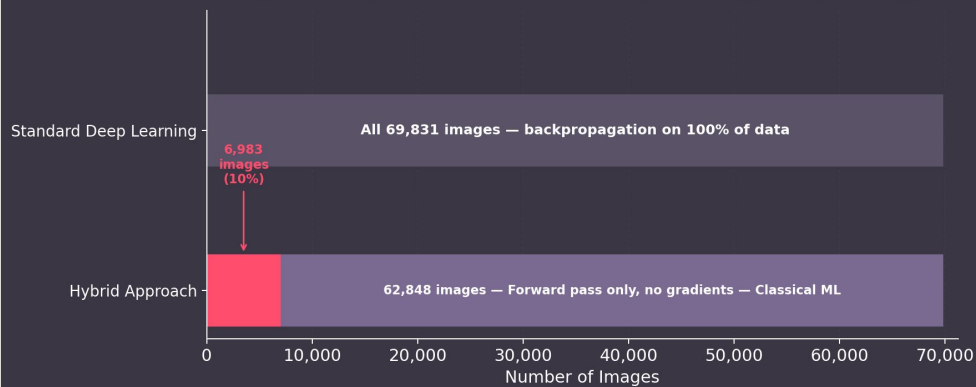
Results:

8

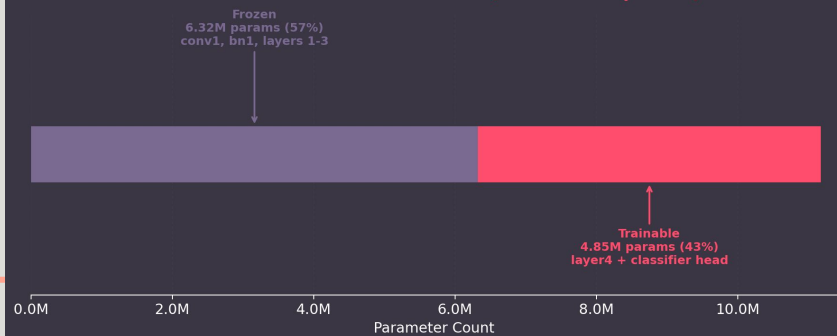


Take away:

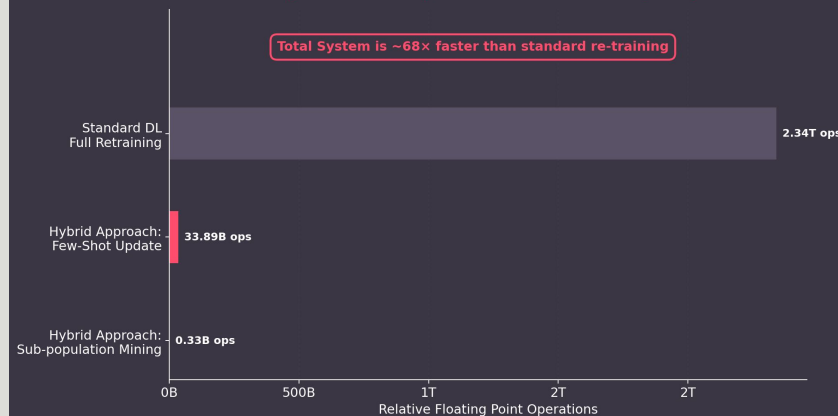
Training Data Requirement: Standard Deep Learning vs. Hybrid Approach



ResNet-18 Parameter Breakdown (11.18M total params)



Operational Multiplier: Scientific Time Complexity



Resources:

- <https://www.merriam-webster.com/dictionary/deepfake>
- <https://www.youtube.com/watch?v=Dp5YwZCGpm0>
- <https://zenodo.org/records/16140829>
- <https://apnews.com/article/pentagon-explosion-misinformation-stock-market-ai-96f534c790872fde67012ee81b5ed6a4>
- <https://www.cnn.com/2024/02/04/asia/deepfake-cfo-scam-hong-kong-intl-hnk>
- <https://www.nbcnews.com/tech/tech-news/musks-ai-chatbot-grok-xai-making-sexual-deepfakes-imagine-rcna265855>